

①

 $\$a_0 = a$ $\$a_1 = b$

check:

ifne $\$a_0, \a_1 , else-case

if-case:

li $\$v_0, 1$

j exit-check

else-case:

li $\$v_0, 0$

exit-check

jr $\$v_0$

②

sum = 0 i = 0 until v_1 & v_2 li $\$v_0, 0$ li $\$t_0, 0$

Loop:

org $\$t_0, \v_1 , endloopsll $\$t_2, \$t_0, 4$ add $\$t_2, \$t_2, \$v_2$ lw $\$t_3, 0(\$t_2)$ add $\$v_0, \$v_0, \$t_3$ addi $\$t_0, \$t_0, 1$

j loop

endloop:

④

sll $\$t_0, \$s_2, 2$ $t_0 = i * 4$ add $\$t_0, \$s_6, \$t_0$ $t_0 = \&A[i]$ lw $\$t_1, 0(\$t_0)$ $t_1 = A[i]$ sub $\$s_0, \$s_1, \$t_1$ $q = h - A[i]$ $q = h - A[i];$ $q = v_0$ $h = v_1$ $i = v_2$ $\&A = v_6$

⑤

$i = x_0$ $j = x_1$ $2A = x_2$ $2B = x_3$

```
addu $t0, $x1, -2 # [j-2]
sll $t0, $t0, 2
add $t0, $x0, $t0
sw $t1, 0($t0)
addi $t1, $t1, 5
sll $t2, $t2, 4
add $t2, $x2, $t2
sw $t1, 0($t2)
```

⑥

```
li $t1, 0 i=0
LOOP:
lw $s1, 0($s0) s1 = &MemArray
add $s2, $s2, $s1 s2 = result + s1
addi $s0, $s0, 4
addi $t1, $t1, 1 i++
slti $t2, $t1, 100 t2 = i < 100
bne $t2, $zero, LOOP
```

int i=0;

for (i=0; i < 100; i++) {

result += MemArray[i];

}

$i = t_1$

result = s_2

&MemArray = x_0

⑦

```
move $s1, $s2 i = n
LOOP:
blez $s1, END_LOOP
sll $t0, $s1, 2 t0 = i * 4
add $t1, $s0, $t0 t1 = &Values[i]
lw $t2, 0($t1) t2 = Values[i]
sll $t2, $t2, 1
sw $t2, 0($t1)
addi $s1, $s1, -1 i--
j LOOP
END_LOOP:
```

for (i = n; i > 0; i--) {

Values[i] = Values[i] * 2;

}

$i = t_1$

$n = x_2$

&Values = x_0

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```

    slt $t0, $r0, 2
    add $t0, $t0, $r2
    lw $t1, 0($t0)
    slt $t2, $r1, 2
    add $t2, $t2, $r3
    lw $t3, 0($t2)
    add $t0, $t0, $r4
    lw $t4, 0($t0)
    add $t4, $t1, $t3
    sw $t5, 0($t4)
  
```

$i = r_0$ $j = r_1$ $\&A = r_2$ $\&B = r_3$ $\&C = r_4$

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```

    addi $t0, $s2, 4  $t_0 = i + 4$ 
    sll $t0, $t0, 2  $t_0 = t_0 * 4$ 
    add $t0, $s6, $t0  $t_0 = A[i + 4]$ 
    lw $t1, 0($t0)  $t_1 = t_0$ 
    sub $s0, $t1, $s1  $g = A[i + 4] - h$ 
  
```

$g = A[i + 4] - h;$

$g = r_0$
 $h = r_1$
 $i = r_2$
 $\&A = r_6$

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```

    LOOP: addi $s1, $zero, -1  $i = -1$ 
           addi $s1, $s1, 1  $i++$ 
           sll $t0, $s1, 2  $t_0 = i * 4$ 
           add $t1, $s0, $t0  $t_1 = \&A[i]$ 
           lw $t2, 0($t1)  $t_2 = t_1$ 
           beq $t2, $zero, LOOP
  
```

```

    i = -1;
    do {
        i++;
    } while (A[i] != 0);
  
```

$i = r_1$
 $\&A = r_0$

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$i = x_0$ $n = x_1$ $2A = x_2$ $2B = x_3$

lw $\$x_0, 0$

Loop:

bge $\$x_0, \$x_1, \text{endLoop}$

sub $\$t_0, \$x_1, \$x_0$

addi $\$t_0, \$t_0, -1$

sll $\$t_1, \$t_0, 2$

add $\$t_1, \$t_1, \$x_2$

lw $\$t_2, 0(\$t_1)$

sll $\$t_3, \$x_0, 2$

add $\$t_3, \$t_3, \$x_3$

sw $\$t_2, 0(\$t_3)$

addi $\$x_0, \$x_0, 1$

j Loop

endLoop: